



Department of Electrical & Computer Engineering (ECE)
Rajshahi University of Engineering & Technology (RUET)

Course Name: Data Base Systems Sessional

Course Code: ECE 2216

Lab Report – 1

Experiment Name:

Creating and Managing Databases using Data Definition Language
(DDL)
and Data Manipulation Language (DML)

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July 14, 2026

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Name of the Experiment

Creating and Managing Databases using Data Definition Language (DDL) and Data Manipulation Language (DML).

Theory

Data Definition Language (DDL)

DDL is the short name for Data Definition Language, which deals with database schemas and descriptions of how the data should reside in the database. It is used to create and modify the structure of database objects.

Data Manipulation Language (DML)

DML focuses on manipulating the data stored in the database, enabling users to retrieve, add, update, and delete data.

Data Query Language (DQL)

DQL is a subset of SQL used to retrieve data from a database without modifying it. Its main command is **SELECT**, which allows users to fetch specific information based on their requirements.

Important SQL Commands Used

- **CREATE (DDL):** Used to create a database and its objects such as tables, indexes, views, stored procedures, functions, and triggers.

```
1 CREATE DATABASE database_name;  
2  
3 CREATE TABLE table_name(  
4     column1_name data_type constraint,  
5     column2_name data_type constraint,  
6     column3_name data_type  
7 );
```

- **INSERT (DML):** Used to insert data into a table.

```
1 INSERT INTO table_name (column1, column2, column3)  
2 VALUES (value1, value2, value3);
```

- **SELECT (DQL):** Used to fetch specific information based on requirements.

```
1 SELECT * FROM table_name;
```

- **ALTER (DDL):** Alters the structure of an existing database/table.
- **RENAME (DDL):** Renames an object.

```

1 ALTER TABLE old_table_name RENAME TO new_table_name;
2 ALTER TABLE table_name RENAME COLUMN old_name TO new_name;

```

- **CHANGE / MODIFY (DDL):** Changes column name or data type.

```

1 ALTER TABLE table_name CHANGE old_column_name new_column_name
  DATA_TYPE;
2 ALTER TABLE table_name MODIFY COLUMN column_name
  NEW_DATA_TYPE;

```

- **ADD (DDL):** Adds a new column.

```

1 ALTER TABLE table_name ADD column_name data_type;

```

- **COMMENT (DDL):** Adds comments to the data dictionary.

```

1 /* this is a comment */

```

- **DELETE (DML):** Deletes records from a table.

```

1 DELETE FROM table_name WHERE condition;

```

- **DROP (DDL):** Deletes objects from the database.

```

1 ALTER TABLE table_name DROP COLUMN column_name;
2 DROP TABLE IF EXISTS users;

```

- **TRUNCATE (DDL):** Removes all records from a table, including allocated storage space.

```

1 TRUNCATE TABLE table_name;

```

Given Dataset: Student Info

Student_ID	Name	Gender	Dept_Name	CGPA	Email
101	Rahim	Male	CSE	3.85	rahim@gmail.com
102	Karim	Male	EEE	3.65	karim@gmail.com
103	Nadia	Female	BBA	3.92	nadia@gmail.com
104	Rina	Female	CSE	3.55	rina@gmail.com
105	Sohan	Male	ME	3.70	sohan@gmail.com

Table 1: Student_Info Dataset

Lab Work

Task 1: Create the Student_Info Table

Explanation: The CREATE TABLE statement (a DDL command) is used to define a new table named Student_Info along with its columns and their respective data types.

```
1 CREATE TABLE Student_Info (  
2     Student_ID INT PRIMARY KEY,  
3     Name VARCHAR(50),  
4     Gender VARCHAR(10),  
5     Dept_Name VARCHAR(20),  
6     CGPA DECIMAL(3,2),  
7     Email VARCHAR(50)  
8 );
```

Listing 1: SQL Code for Task 1

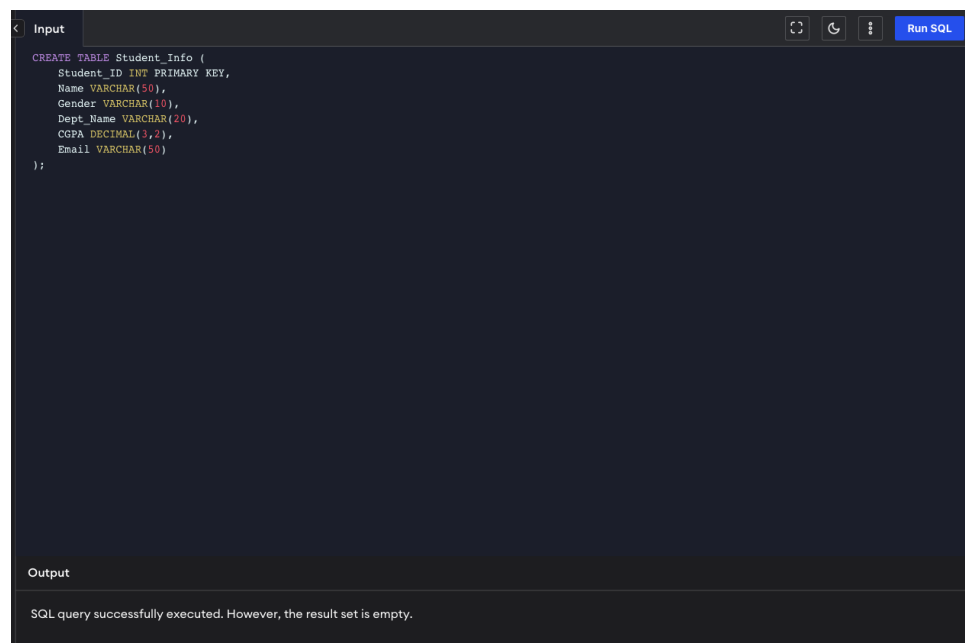


Figure 1: Output of Task 1

Task 2: Insert Records into Student_Info

Explanation: The INSERT INTO statement (a DML command) is used to insert all five given student records into the Student_Info table.

```
1 INSERT INTO Student_Info (Student_ID, Name, Gender, Dept_Name,  
2     CGPA, Email)  
3 VALUES  
4 (101, 'Rahim', 'Male', 'CSE', 3.85, 'rahim@gmail.com'),  
5 (102, 'Karim', 'Male', 'EEE', 3.65, 'karim@gmail.com'),  
6 (103, 'Nadia', 'Female', 'BBA', 3.92, 'nadia@gmail.com'),  
7 (104, 'Rina', 'Female', 'CSE', 3.55, 'rina@gmail.com'),  
8 (105, 'Sohan', 'Male', 'ME', 3.70, 'sohan@gmail.com');
```

Listing 2: SQL Code for Task 2

The screenshot shows the PostgreSQL interface with the following SQL code in the input field:

```
SELECT * FROM Student_Info (Student_ID, Name, Gender, Dept_Name, CGPA, Email);
```

The output area displays the message: "SQL query successfully executed. However, the result set is empty."

On the right side, the 'Available Tables' section lists the following tables:

- Customers**: customer_id, first_name, last_name, age, country
- Orders**: order_id, item, amount, customer_id
- Shipments**: shipping_id, status, customer
- Student_Info**: Student_ID, Name, Gender, Dept_Name, CGPA, Email

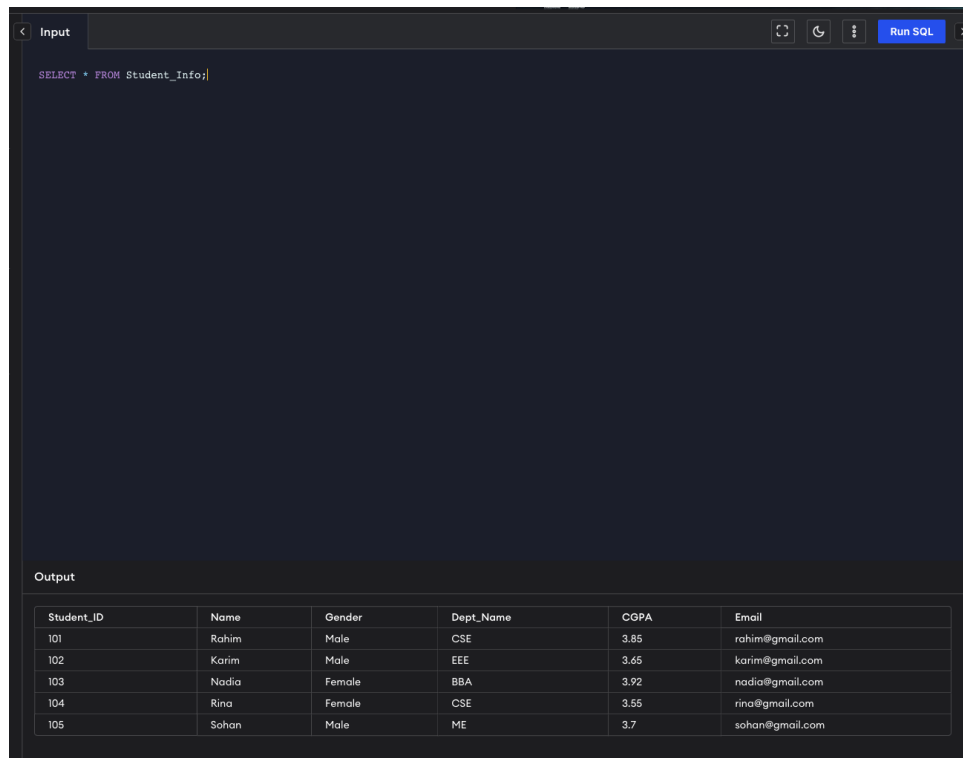
Figure 2: Output of Task 2

Task 3: Display All Records

Explanation: The `SELECT *` statement (a DQL command) retrieves and displays all columns and rows from the `Student_Info` table.

```
1 SELECT * FROM Student_Info;
```

Listing 3: SQL Code for Task 3



Input

```
SELECT * FROM Student_Info;
```

Output

Student_ID	Name	Gender	Dept_Name	CGPA	Email
101	Rahim	Male	CSE	3.85	rahim@gmail.com
102	Karim	Male	EEE	3.65	karim@gmail.com
103	Nadia	Female	BBA	3.92	nadia@gmail.com
104	Rina	Female	CSE	3.55	rina@gmail.com
105	Sohan	Male	ME	3.7	sohan@gmail.com

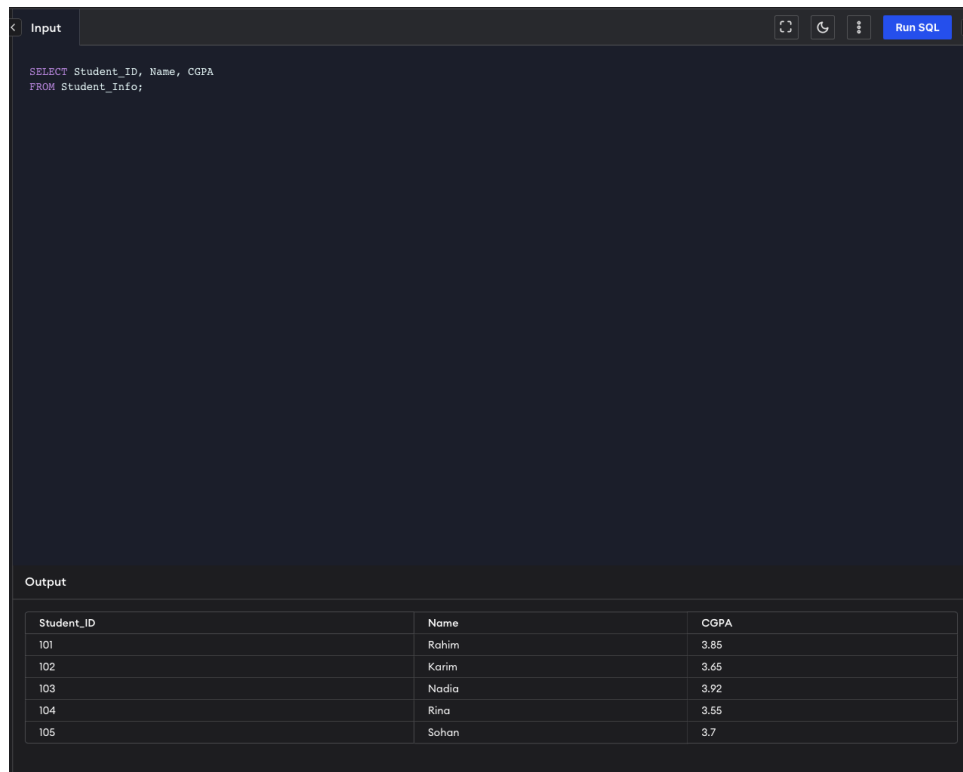
Figure 3: Output of Task 3

Task 4: Display Student_ID, Name, and CGPA Only

Explanation: A SELECT statement with specific column names is used to retrieve only the required columns (Student_ID, Name, CGPA) instead of the entire table.

```
1 SELECT Student_ID, Name, CGPA
2 FROM Student_Info;
```

Listing 4: SQL Code for Task 4



The screenshot shows a SQL IDE interface. The top bar has a tab labeled 'Input' and a 'Run SQL' button. The main area contains the following SQL query:

```
SELECT Student_ID, Name, CGPA
FROM Student_Info;
```

Below the query, the 'Output' section displays a table with the following data:

Student_ID	Name	CGPA
101	Rahim	3.85
102	Karim	3.65
103	Nadia	3.92
104	Rina	3.55
105	Sohan	3.7

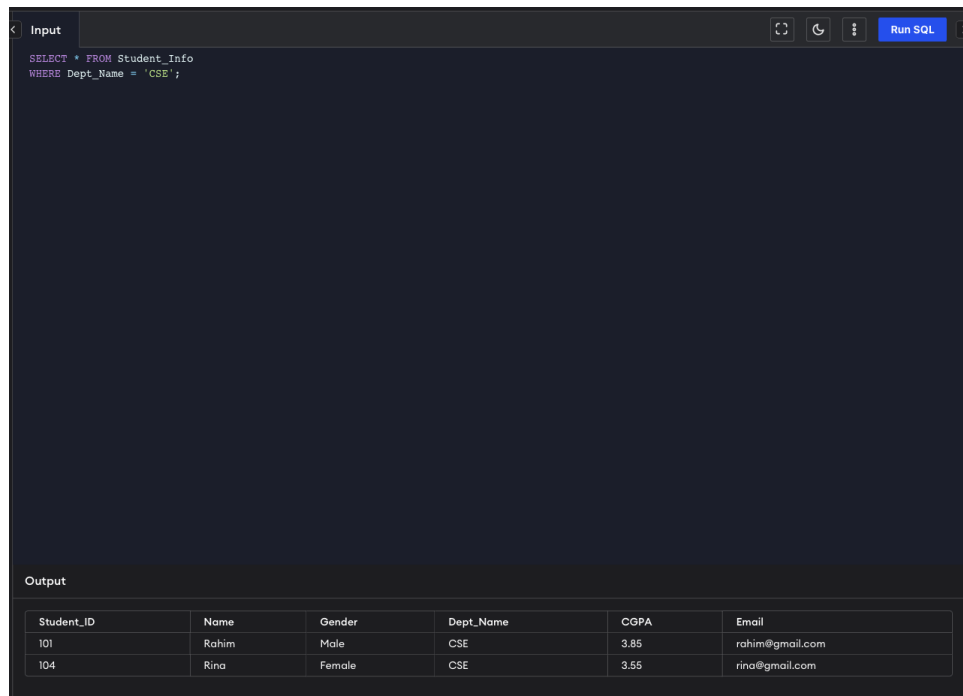
Figure 4: Output of Task 4

Task 5: Display Information of CSE Department Students

Explanation: The WHERE clause is used along with SELECT to filter and display only the records where Dept_Name equals 'CSE'.

```
1 SELECT * FROM Student_Info
2 WHERE Dept_Name = 'CSE';
```

Listing 5: SQL Code for Task 5



The screenshot shows a SQL IDE interface. The top bar has a tab labeled 'Input' and a 'Run SQL' button. The main area contains the following SQL query:

```
SELECT * FROM Student_Info  
WHERE Dept_Name = 'CSE';
```

Below the query, the 'Output' section displays a table with the following data:

Student_ID	Name	Gender	Dept_Name	CGPA	Email
101	Rahim	Male	CSE	3.85	rahim@gmail.com
104	Rina	Female	CSE	3.55	rina@gmail.com

Figure 5: Output of Task 5

Task 6: Change Sohan's Department from ME to EEE

Explanation: The UPDATE statement (a DML command) modifies existing records. Here, it changes the Dept_Name of the student named Sohan from 'ME' to 'EEE'.

```
1 UPDATE Student_Info  
2 SET Dept_Name = 'EEE'  
3 WHERE Name = 'Sohan';
```

Listing 6: SQL Code for Task 6

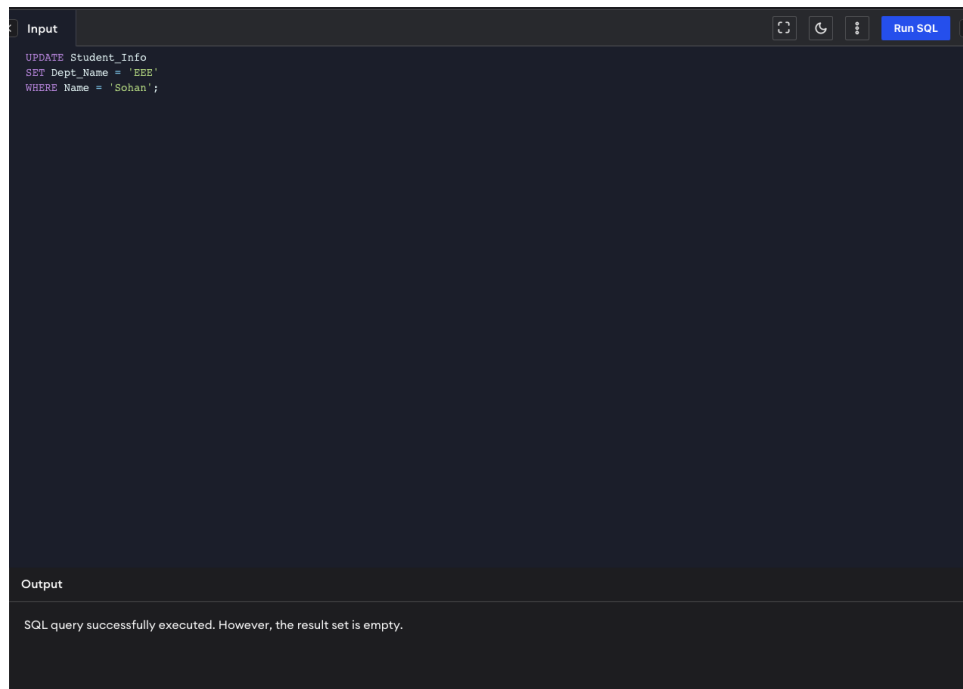


Figure 6: Output of Task 6

Task 7: Delete Record of Student_ID = 104

Explanation: The DELETE statement (a DML command) removes the record of the student whose Student_ID is 104, i.e., Rina, from the Student_Info table.

```

1 DELETE FROM Student_Info
2 WHERE Student_ID = 104;

```

Listing 7: SQL Code for Task 7

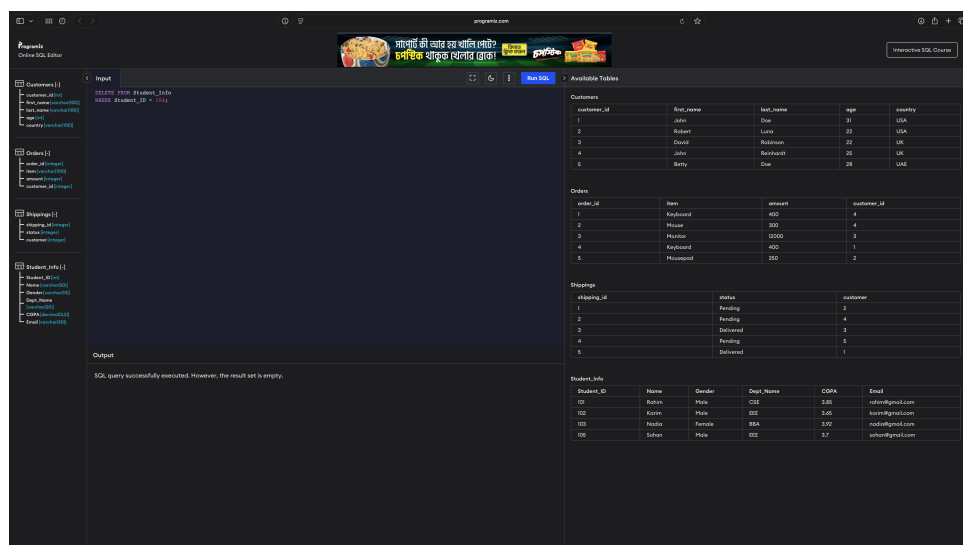


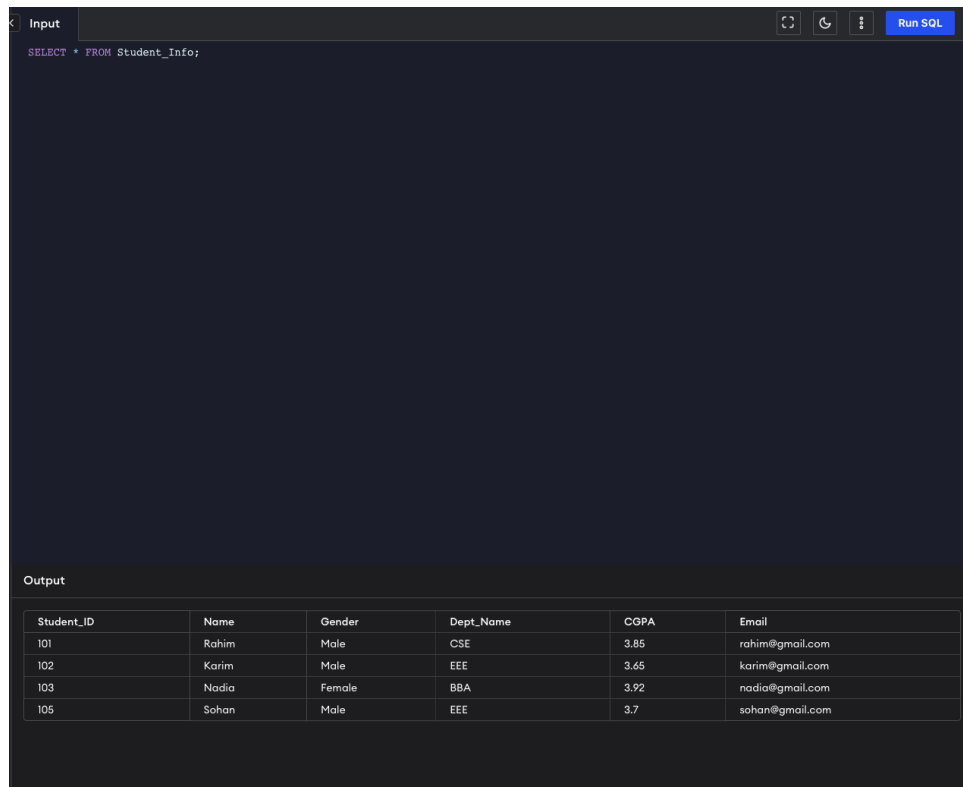
Figure 7: Output of Task 7

Task 8: Display Final Contents of the Table

Explanation: A final `SELECT *` query is executed to verify and display the updated contents of the `Student_Info` table after the update and delete operations.

```
1 SELECT * FROM Student_Info;
```

Listing 8: SQL Code for Task 8



The screenshot shows a SQL IDE interface. The 'Input' tab at the top contains the query `SELECT * FROM Student_Info;`. Below the input area, the 'Output' tab displays a table with the following data:

Student_ID	Name	Gender	Dept_Name	CGPA	Email
101	Rahim	Male	CSE	3.85	rahim@gmail.com
102	Karim	Male	EEE	3.65	karim@gmail.com
103	Nadia	Female	BBA	3.92	nadia@gmail.com
105	Sohan	Male	EEE	3.7	sohan@gmail.com

Figure 8: Output of Task 8

Discussion

In this lab, we practically implemented the fundamental concepts of Data Definition Language (DDL), Data Manipulation Language (DML), and Data Query Language (DQL) using SQL. We began by creating a table named `Student_Info` with appropriate columns and data types using the `CREATE TABLE` statement. We then inserted five student records into the table using the `INSERT INTO` statement.

Using the `SELECT` statement, we retrieved data in different ways: viewing the entire table, viewing only selected columns (`Student_ID`, `Name`, `CGPA`), and filtering records based on a condition (`department = CSE`) using the `WHERE` clause. We also modified an existing record using the `UPDATE` statement, changing a student's department, and removed a specific record using the `DELETE` statement.

Through this exercise, we gained hands-on experience with the basic building blocks of database management: creating schemas, inserting and modifying data, and querying information based on specific conditions. This lab strengthened our understanding of how DDL, DML, and DQL commands work together to manage a relational database effectively.